

Maze Solving. We leverage the maze-dataset package^{§§} to generate 2D unsolved mazes along with their corresponding ground truth solutions. We use "circle" shape to denote the start position and "star" shape to denote the end position. We generate two subsets featuring 2×2 and 3×3 maze configurations.

L Dataset Statistics

		# Training Instances	# Eval Instances
Probing Tasks	Line or Angle	10K	1K
	Angle Classification	10K	1000
	Length Comparison	10K	1000
	Clevr QA	36K	1000
	Shapeworld Scene	15K	100
	Maze Scene	10K	600
Zero-Shot Downstream Tasks	Angle Classification	-	100
	Length Comparison	-	100
	Shapeworld Spatial Reasoning (2Obj)	-	100
	Shapeworld Spatial Reasoning (MultiObj)	-	100
	Shapeworld Superlative	-	100
	NLVR	-	100
	Geoclidean 2-shot Learning	-	100
	2×2 Maze Solving	-	100
	3×3 Maze Solving	-	100
	VGBench-QA Category	-	100
	VGBench-QA Color	-	100
	VGBench-QA Usage	-	100

Table 13: Statistics of the probing tasks (§ A.1) and the downstream tasks (§ 3). The GPT-4(V) zero-shot results on probing tasks are reported on 100 randomly sub-sampled instances from the entire eval split.

Detailed statistics of the probing tasks used in § A and the zero-shot downstream tasks mentioned in § 3 can be found in Table 13.

^{§§}<https://github.com/understanding-search/maze-dataset/tree/main>

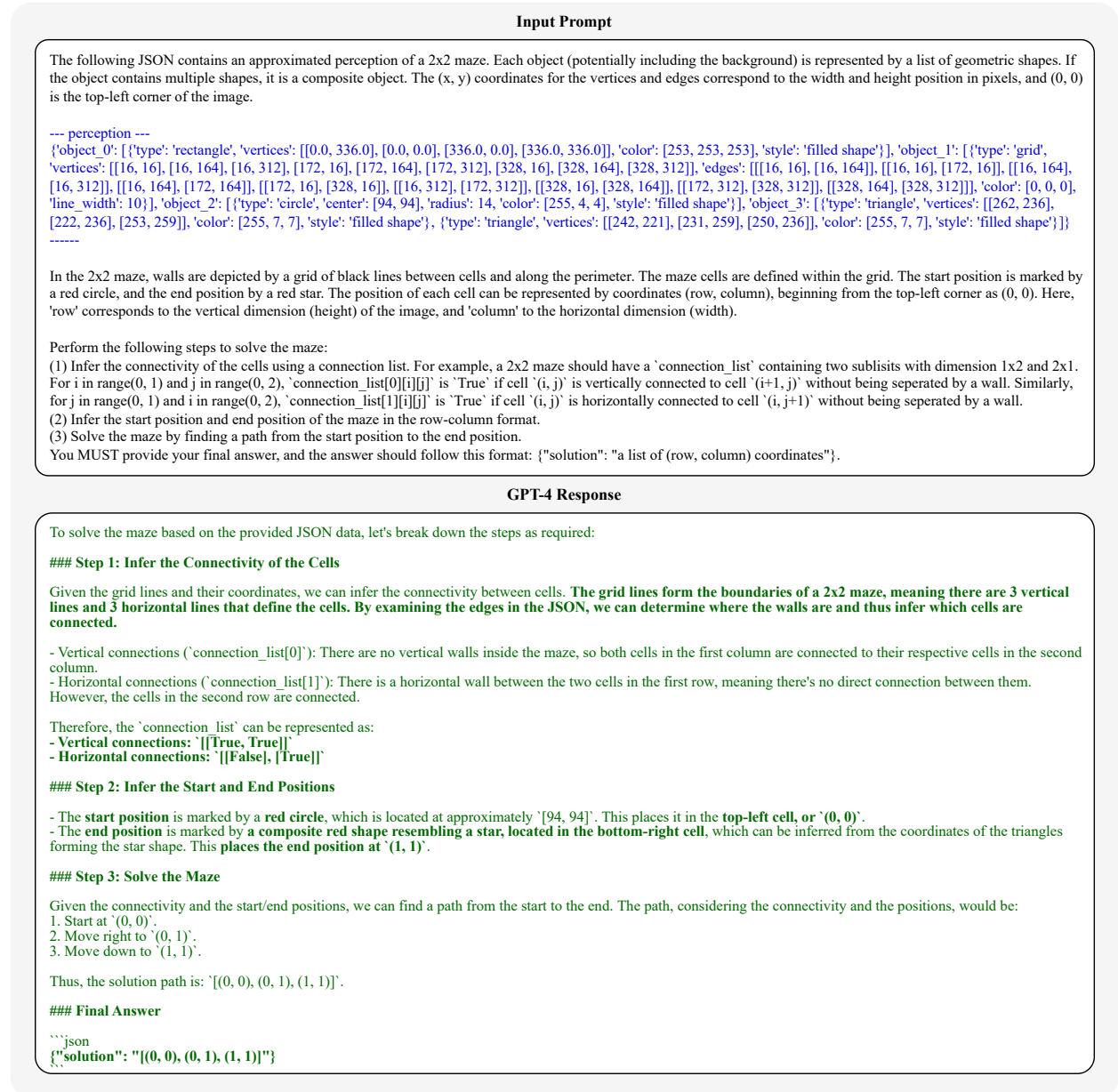


Figure 17: Full input prompt and GPT-4 response of the 2x2 maze solving example in Figure 2. The blue part in the input prompt indicates the generated Primal Visual Description (PVD) of the entire image.

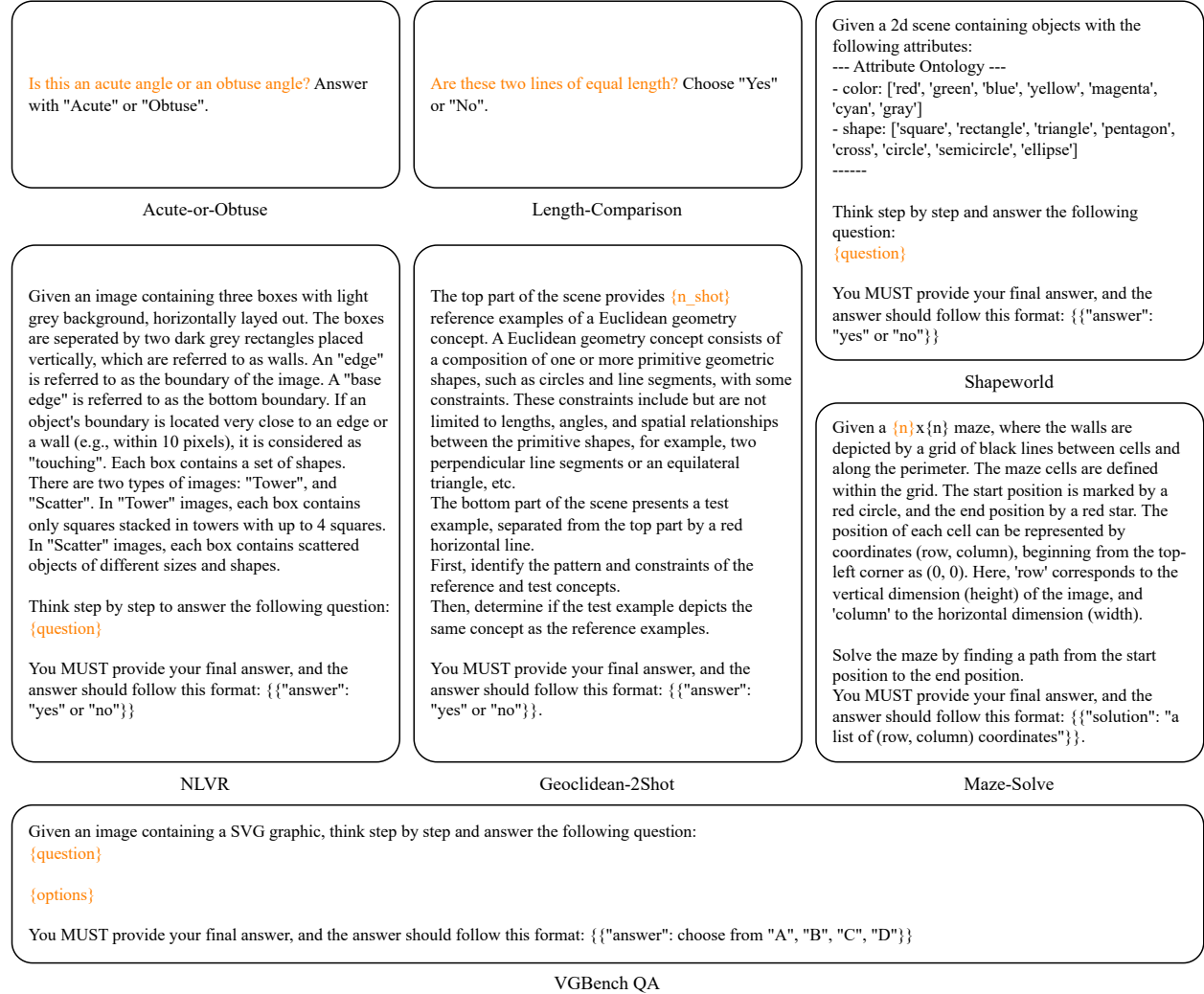


Figure 18: Prompts for zero-shot downstream tasks with image input

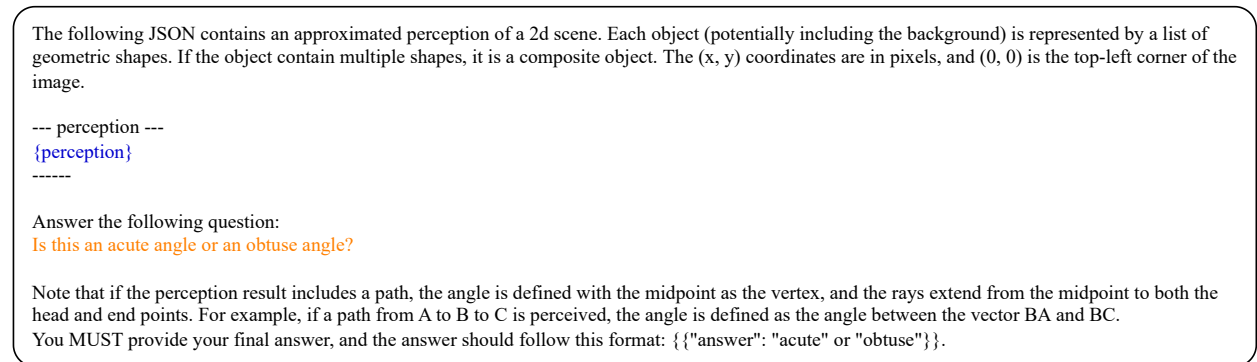


Figure 19: Prompt for task Angle Classification with Primal Visual Description perception input.